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5 RCE in Models with Heterogeneous Agents

This section is based on models with idiosyncratic shocks by Bewley (1983), Huggett (1993), Aiyagari (1994), and Krusell and Smith (1998) (see also Chapter 17 from Ljungqvist and Sargent (2012)).

Environment

- Continuum of measure one of agents
- Preferences: $E \sum_{t=0}^{\infty} \beta^t U(c_t)$
- Income endowment process y_t —Markov process with the transition function Q
- The agents can save and borrow at rate r
- Let a denote the asset holdings
- Borrowing limits: $a \geq \underline{a}$, where $\underline{a} < 0$
- The individual state variables are a and y

First, suppose that r is constant over time. Then the individual's problem is

$$v(a, y) = \max_{\underline{a} \leq a' \leq (1+r)a+y} \left\{ U((1+r)a + y - a') + \beta \int v(a', y') Q(y, dy') \right\}$$

Now, let's put this into an equilibrium setting. There is a mass one of agents. Each agent faces the same stochastic process (common Q), but realizations are i.i.d. across individuals.

In this case, r will be determined endogenously. In particular, in a RCE, r will be a function of the aggregate state variable.

The aggregate state variable is the distribution of individuals across states: $\lambda(a, y)$.

Recall that in standard growth model with the representative agent, the aggregate state variable was K . This is because the distribution is degenerate, and hence K contains all the relevant information.

Recursive Competitive Equilibrium: Definition

A RCE is a list of functions

- $v(a, y, \lambda)$ - value function
- $g(a, y, \lambda)$ - policy function
- $r(\lambda)$
- $H(\lambda)$ - next period's distribution λ'

(1) Consumer maximization

Taking $r(\lambda)$ and $H(\lambda)$ as given, $v(a, y, \lambda)$ solves

$$v(a, y, \lambda) = \max_{b \leq a' \leq (1+r(\lambda))a+y} \left\{ U((1+r(\lambda))a + y - a') + \beta \int v(a', y', H(\lambda)) Q(y, dy') \right\}$$

and $g(a, y, \lambda)$ is the optimal decision rule.

(2) Consistency

Given λ , the next period's distribution $\lambda' = H(\lambda)$ is consistent with the transition function Q and the decision rule $g(a, y, \lambda)$.

(3) Market clearing

- Asset market: $\int g(a, y, \lambda) d\lambda = 0$ for all feasible λ .
- Goods market clears by Walras' law: $\int [(1+r(\lambda))a + y - g(a, y, \lambda)] d\lambda = \int y d\lambda$ for all feasible λ .

Understanding Consistency

To understand what part (2) of the definition is, and what H refers to, consider for simplicity a special case, discretization.

Suppose that there is a finite number of income levels, $\{\bar{y}_1, \bar{y}_2, \dots, \bar{y}_N\}$, and a finite number of asset levels, $\{\bar{a}_1, \bar{a}_2, \dots, \bar{a}_M\}$, $\bar{a}_i < \bar{a}_{i+1}$, $\bar{a}_1 = \underline{b}$. Now λ is an $N \times M$ matrix, where the columns correspond to $\bar{a}_j$'s and the rows correspond to $\bar{y}_i$'s. I.e., λ_{ij} is the mass of households with income $\bar{y}_i$ and assets $\bar{a}_j$.

In other words, at time t λ_t is the unconditional distribution of (a_t, y_t) pairs: $\lambda_t(a, y) = \Pr(a_t = a, y_t = y)$.

The law of motion H is given by

$$\begin{aligned} \underbrace{\Pr(a_{t+1} = a', y_{t+1} = y')}_{\lambda'} &= \sum_{a_t} \sum_{y_t} \Pr(a_{t+1} = a', y_{t+1} = y' | a_t = a, y_t = y) \Pr(a_t = a, y_t = y) \\ &= \sum_{a_t} \sum_{y_t} \underbrace{\Pr(a_{t+1} = a' | a_t = a, y_t = y)}_{=1 \text{ if } g(a, y, \lambda) = a', =0 \text{ o/w}} \underbrace{\Pr(y_{t+1} = y' | y_t = y)}_{Q(y, y')} \underbrace{\Pr(a_t = a, y_t = y)}_{\lambda} \end{aligned}$$

That is,

$$\lambda'(a', y') = \sum_a \sum_y \mathcal{I}(a', a, y, \lambda) Q(y, y') \lambda(a, y)$$

where
$$\begin{cases} \mathcal{I}(a', a, y, \lambda) = 1 & \text{if } a' = g(a, y, \lambda), \\ \mathcal{I}(a', a, y, \lambda) = 0 & \text{otherwise,} \end{cases}$$

which can be rewritten as

$$\lambda'(a', y') = \sum_y \sum_{a: a' = g(a, y, \lambda)} Q(y, y') \lambda(a, y)$$

Then in our definition, $H(\lambda) = \lambda'$ is defined by the above equation.

If a and y are not discrete, then we have

$$\lambda'(a', Y') = \int_y \int_{a: a' = g(a, y, \lambda)} Q(y, Y') d\lambda$$

where Y' is a Borel set for the state variable y .

Discrete case

You can derive a linear mapping from λ to λ' : $\lambda' = B\lambda$.

Suppose $M = N = 2$, and let the decision rule $g(\bar{y}, \bar{a})$ be as follows: $g(\bar{y}_1, \bar{a}_1) = \bar{a}_1$, $g(\bar{y}_1, \bar{a}_2) = \bar{a}_2$, $g(\bar{y}_2, \bar{a}_1) = \bar{a}_2$, $g(\bar{y}_2, \bar{a}_2) = \bar{a}_1$.

i.i.d. case: $Pr(y = \bar{y}_1) = \pi$

Therefore, individuals in state $(i, j) \equiv (\bar{y}_i, \bar{a}_j)$ have the following probabilities to be in tomorrow's states:

$$(1, 1) \rightarrow \begin{cases} (1, 1), & \text{w/ } \pi \\ (2, 1), & \text{w/ } 1 - \pi \end{cases}$$

$$(1, 2) \rightarrow \begin{cases} (1, 2), & \text{w/ } \pi \\ (2, 2), & \text{w/ } 1 - \pi \end{cases}$$

$$(2, 1) \rightarrow \begin{cases} (1, 2) & \text{w/ } \pi \\ (2, 2) & \text{w/ } 1 - \pi \end{cases}$$

$$(2, 2) \rightarrow \begin{cases} (1, 1) & \text{w/ } \pi \\ (2, 1) & \text{w/ } 1 - \pi \end{cases}$$

This gives the following system of equations which implicitly defines the B matrix

$$\begin{cases} \lambda'_{11} = \pi \lambda_{11} + \pi \lambda_{22} \\ \lambda'_{12} = \pi \lambda_{12} + \pi \lambda_{21} \\ \lambda'_{21} = (1 - \pi) \lambda_{11} + (1 - \pi) \lambda_{22} \\ \lambda'_{22} = (1 - \pi) \lambda_{12} + (1 - \pi) \lambda_{21} \end{cases}$$

Thus, we have

$$\begin{bmatrix} \lambda'_{11} \\ \lambda'_{12} \\ \lambda'_{21} \\ \lambda'_{22} \end{bmatrix} = \begin{bmatrix} \pi & 0 & 0 & \pi \\ 0 & \pi & \pi & 0 \\ 1 - \pi & 0 & 0 & 1 - \pi \\ 0 & 1 - \pi & 1 - \pi & 0 \end{bmatrix} \begin{bmatrix} \lambda_{11} \\ \lambda_{12} \\ \lambda_{21} \\ \lambda_{22} \end{bmatrix}$$

so in this case $\lambda' = H(\lambda) = B\lambda$.

Markov w/ persistence π case: Now suppose $Pr(y' = \bar{y}_i | y = \bar{y}_i) = \pi$, $i = 1, 2$.

Then

$$(1, 1) \rightarrow \begin{cases} (1, 1), & \text{w/ } \pi \\ (2, 1), & \text{w/ } 1 - \pi \end{cases}$$

$$(1, 2) \rightarrow \begin{cases} (1, 2), & \text{w/ } \pi \\ (2, 2), & \text{w/ } 1 - \pi \end{cases}$$

$$(2, 1) \rightarrow \begin{cases} (1, 2) & w/ 1 - \pi \\ (2, 2) & w/ \pi \end{cases}$$

$$(2, 2) \rightarrow \begin{cases} (1, 1) & w/ 1 - \pi \\ (2, 1) & w/ \pi \end{cases}$$

so that

$$\begin{cases} \lambda'_{11} = \pi \lambda_{11} + (1 - \pi) \lambda_{12} \\ \lambda'_{12} = \pi \lambda_{21} + (1 - \pi) \lambda_{22} \\ \lambda'_{21} = (1 - \pi) \lambda_{11} + \pi \lambda_{12} \\ \lambda'_{22} = (1 - \pi) \lambda_{21} + \pi \lambda_{22} \end{cases}$$

Thus, we have

$$\begin{bmatrix} \lambda'_{11} \\ \lambda'_{12} \\ \lambda'_{21} \\ \lambda'_{22} \end{bmatrix} = \begin{bmatrix} \pi & 0 & 0 & 1 - \pi \\ 0 & \pi & 1 - \pi & 0 \\ 1 - \pi & 0 & 0 & \pi \\ 0 & 1 - \pi & \pi & 0 \end{bmatrix} \begin{bmatrix} \lambda_{11} \\ \lambda_{12} \\ \lambda_{21} \\ \lambda_{22} \end{bmatrix}$$

As a practical matter, solving for the RCE can be difficult. Why?

1. We cannot solve a SP problem and back out equilibrium prices, because markets are incomplete and hence welfare theorems do not apply.
2. $r(\cdot)$ is a function from measures into real numbers, so the price function is a very high-dimensional object

Even though solving for equilibrium is difficult, solving for a steady-state equilibrium is not so hard.

Steady-State Equilibrium, or Stationary Recursive Competitive Equilibrium (SRCE)

A SRCE is a RCE and λ , call it λ^* , such that $\lambda^* = H(\lambda^*)$. The aggregate, the interest rate, consumption etc. are constant, but things are changing at the individual level.

Interpretation of $\lambda^(a, y)$:*

- Fraction of time spent by an infinitely-lived HH in state (a, y) .
- Fraction of HHs in state (a, y) at any given period.

Although each HH changes its state over time, the aggregate distribution remains unchanged.

An alternative (shorter) definition of a stationary equilibrium in this case

A SRCE is a list of functions $v(a, y)$, $g(a, y)$, r , $\lambda(a, y)$ such that

1. Given r , $v(a, y)$ solves

$$v(a, y) = \max_{b \leq a' \leq (1+r)a+y} \left\{ U((1+r)a + y - a') + \beta \int_{y'} v(a', y') Q(y, dy') \right\}$$

and $g(a, y)$ is the optimal decision rule.

2. The stationary distribution λ is induced by Q and $g(a, y)$
3. The loan markets clears:

$$\int g(a, y) d\lambda = 0.$$

Algorithm for Finding a Steady State Equilibrium

The key point is that in steady state (SS) prices are constant because $\lambda = \lambda^*$ such that $\lambda^* = H(\lambda^*)$, and the problem of finding SS can be thought of as finding the SS value of r . The algorithm is

1. Guess r .
2. Compute $v(a, y)$ and $g(a, y)$.
3. Given $g(a, y)$ and $Q(y, y')$, we can construct the mapping H .
4. Then guess λ_0 , find $\lambda_1 = H(\lambda_0)$, $\lambda_2 = H(\lambda_1)$, ... and iterate until convergence (find the fixed point of H : $\lim_{t \rightarrow \infty} \lambda_t = \lambda^*$).
5. Compute $\int g(a, y) d\lambda^*$. If this equals zero, we are done.
 - if $< 0 \Rightarrow$ increase r , go to step 2.

- if $> 0 \Rightarrow$ decrease r , go to step 2.

Extending the Analysis to a Setting Like the Growth Model

Examples of this type of setup are Huggett (1993), Aiyagari (1994), and Krusell and Smith (1998).

Model Setup

- There is a measure one of ex-ante identical individuals with preferences given by $E \sum_{t=0}^{\infty} \beta^t U(c_t, 1 - n_t)$.
- The efficiency of an individual's time devoted to work in period t is denoted by e_{it} . Q is the Markov transition function, which is the same across individuals but realization are i.i.d. across individuals.
- Technology: the aggregate production function is $Y_t = F(K_t, N_t)$, where $N_t = \int_0^1 n_{it} e_{it} di$ and F is CRS.
- Endowments: each individual is endowed with one unit of time each period, and individual i has $k_{i,0}$ units of capital at time zero. The initial values for the efficiency units are $e_{i,0}$.
- Incomplete markets
 - In each period, there is a market for output, and markets for capital and labor services.
 - There are no markets for insurance (i.e. state-contingent claims), and no opportunities to borrow.

State variables

The state variables of the RCE are

- Individual:
 - Household: e, k .
 - Firm: none.
- Aggregate: $\lambda(e, k)$.

RCE definition

A RCE is a list of functions: $v(e, k, \lambda), n(e, k, \lambda), k'(e, k, \lambda), k^f(\lambda), n^f(\lambda), r(\lambda), w(\lambda), H(\lambda)$ such that

(1) (Consumer Maximization)

Taking $r(\lambda), w(\lambda)$, and $H(\lambda)$ as given, $v(e, k, \lambda)$ solves

$$v(e, k, \lambda) = \max_{n, k'} \left\{ U(r(\lambda)k + w(\lambda)ne + (1-\delta)k - k', 1-n) + \beta \int_{e'} v(e', k', H(\lambda)) Q(e, de') \right\}$$

$$\text{s.t.} \quad r(\lambda)k + w(\lambda)ne + (1-\delta)k - k' \geq 0$$

$$k' \geq 0, \quad 0 \leq n \leq 1,$$

and $n(e, k, \lambda), k'(e, k, \lambda)$ are the corresponding optimal decision rules.

(2) (Firm Maximization)

Taking $r(\lambda), w(\lambda)$ as given, for each λ , $k^f(\lambda)$ and $n^f(\lambda)$ solve

$$\max_{k, n} F(k, n) - r(\lambda)k - w(\lambda)n$$

$$\text{s.t.} \quad k \geq 0, \quad n \geq 0.$$

(3) (Consistency)

Given $\lambda(e, k)$, and given $k'(e, k, \lambda)$ and $Q(e, e')$, the next period's distribution $\lambda' = H(\lambda)$ satisfies

$$\lambda'(E', k') = \int_e \int_{k: k'(e, k, \lambda) = k'} Q(e, E') \lambda(de, dk)$$

for any Borel set E' of the space for the state variable e .

(4) (Market Clearing) For all feasible λ ,

$$\int n(e, k, \lambda) e d\lambda = n^f(\lambda),$$

$$\int k d\lambda = k^f(\lambda).$$

Steady-State Equilibrium

A SPCE for this economy is a RCE and a value of $\lambda = \lambda^* : \lambda^* = H(\lambda^*)$.

Solving for the SS equilibrium is relatively easy. Note that in SS prices are constant, so r and w are just numbers. Also, since F is CRS, e.g. $F(K, N) = K^\alpha N^{1-\alpha}$
 $\Rightarrow$

$$r = \alpha \left(\frac{K}{N} \right)^{\alpha-1}, \quad w = (1 - \alpha) \left(\frac{K}{N} \right)^\alpha.$$

In general, r and w are both functions of K/N so a given value of r implies a particular value for w .

Algorithm

1. Guess r and solve for w .
2. Solve the consumer's Bellman equation with constant r and w to get the decision rules k', n .
3. Solve for the invariant measure λ^* that corresponds to the decision rule $k'(e, k)$ and $Q(e, e')$
4. Check the market clearing for capital: From the firm's problem we only get k^f/n^f . To compute k^f separately, define the aggregate labor supply as $N = \int e n(e, k) d\lambda^*$. Use the fact that in equilibrium labor supply=labor demand, and thus compute $k^f = (k^f/n^f) \times N$. Then check if it equals capital supply, which is $\int k d\lambda^*$. Adjust r accordingly, iterate.

Some observations

Special Case 1: $U(c, 1 - n) = u(c) + v(1 - n)$

In the **deterministic** representative agent model, the optimality condition is

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} = 1 + \underbrace{r_{t+1} - \delta}_{\text{interest rate}},$$

which in a SS yields

$$\frac{u'(c^*)}{\beta u'(c^*)} = 1 + r^* - \delta \Rightarrow 1 + \underbrace{r^* - \delta}_{\text{interest rate}} = \frac{1}{\beta}$$

In the **stochastic** model with incomplete markets and heterogeneous agents, the consumer's FOC is

$$u'(c_t) \geq \beta(1 + r^* - \delta) \int u'(c_{t+1}) dF(c_{t+1})$$

which holds with " $=$ " if $k_{t+1} > 0$ (or $a_{t+1} > \underline{b}$), and with " $>$ " if $k_{t+1} = 0$ (or $a_{t+1} = \underline{b}$).

Integrating both sides across the population,

$$\int u'(c_t) d\lambda^* \geq \beta(1 + r^* - \delta) \int \int u'(c_{t+1}) dF(c_{t+1}) d\lambda^*.$$

Since in SS the distribution of consumption across the population is constant we have

$$1 \geq \beta(1 + r^* - \delta) \Rightarrow 1 + r^* - \delta \leq \frac{1}{\beta},$$

with strict inequality as long as non-zero measure of population is constrained (these people will be the ones experiencing a sequence of adverse shocks).

Thus, r^* will be lower in this economy than in the same economy where we remove all idiosyncratic shocks.

Special Case 2: Leisure is not valued

N^* is now exogenous, an a lower value of r^* implies a higher value of K^*/N^* , and hence K^* . Therefore, the economy with idiosyncratic risk and incomplete markets over-accumulates capital, due to a precautionary savings motive.

Model with Aggregate Production Shocks

Follows Krusell and Smith (1998).

Let us modify Aiyagari (1994) model by adding an aggregate production shock z , that follows a Markov process with the transition function P .

State variables

- Individual:
 - HH: e, k
 - Firm: none
- Aggregate: $\lambda(e, k), z$

RCE Definition

A RCE for this economy is a list of functions

- $v(e, k, \lambda, z)$
- $k^f(\lambda, z)$
- $n(e, k, \lambda, z)$
- $r(\lambda, z)$
- $k'(e, k, \lambda, z)$
- $w(\lambda, z)$
- $n^f(\lambda, z)$
- $H(\lambda, z)$

such that

(1) (Consumer Maximization)

Taking $r(\lambda, z)$, $w(\lambda, z)$, $H(\lambda, z)$ as given, $v(e, k, \lambda, z)$ solves

$$\begin{aligned}
 v(e, k, \lambda, z) = & \max_{n, k'} \left\{ U(r(\lambda, z)k + w(\lambda, z)ne + (1 - \delta)k - k', 1 - n) \right. \\
 & \left. + \beta \int_{z'} \int_{e'} v(e', k', H(\lambda, z), z') Q(e, de') P(z, dz') \right\} \\
 \text{s.t.} \quad & r(\lambda, z)k + w(\lambda, z)ne + (1 - \delta)k - k' \geq 0 \\
 & k' \geq 0, \quad 0 \leq n \leq 1
 \end{aligned}$$

and $n(e, k, \lambda, z)$ and $k'(e, k, \lambda, z)$ are the optimal decision rules.

(2) (Firm Maximization)

For each λ and z , taking $r(\lambda, z)$, $w(\lambda, z)$ as given, $(k^f(\lambda, z), n^f(\lambda, z))$ is a solution to

$$\begin{aligned}
 \max_{k, n} \quad & zF(k, n) - kr(\lambda, z) - nw(\lambda, z) \\
 \text{s.t.} \quad & k \geq 0, \quad n \geq 0
 \end{aligned}$$

(3) (Consistency)

For each λ and z , given $k'(e, k, \lambda, z)$ and Q , the next period's distribution $\lambda' = H(\lambda, z)$ satisfies

$$\lambda'(E', k') = \int_e \int_{k: k' = k'(e, k, \lambda, z)} Q(e, E') \lambda(de, dk)$$

(4) (Market Clearing)

$$\begin{aligned}\int en(e, k, \lambda, z) d\lambda &= n^f(\lambda, z) \\ \int k d\lambda &= k^f(\lambda, z) \\ \int [r(\lambda, z)k + w(\lambda, z)en(e, k, \lambda, z) + k'(e, k, \lambda, z) - (1 - \delta)k] d\lambda \\ &= zF(k^f(\lambda, z), n^f(\lambda, z))\end{aligned}$$

for all feasible λ, z .

This is hard to compute, because the dimension of the state vector increases with the size of the grid (“curse of dimensionality”).

One way to approach this is to characterize λ by a finite set of moments $m = (m_1, m_2, \dots, m_I)$ (See LS Chapter 17 or the original paper for more details).

The main result is that the behavior of the macroeconomics aggregates can be almost perfectly described using only the mean of the wealth distribution and the aggregate productivity shock.

Specifically, the authors compare the aggregate time series K_t, N_t, r_t, w_t from this model with a corresponding representative agent model (or complete markets). They find that the statistics for the aggregate quantities and prices for the two types of models are very close. Hence, fluctuations in the distribution of wealth have only a small effect on the aggregate variables.

Krusell and Smith (1998) compare the distribution of wealth from their model to the U.S. data.

They find

- With a constant discount factor β , the model generates too few very poor people and too many rich people.
- They modify the model so that β is a random variable that can take two values.
- They find that a modest difference between these two values can bring the model’s wealth distribution much closer to the data (patient people becomes

wealthier, impatient eventually becomes poorer).

Heterogeneity Among Firms

This section is based on Davis and Haltiwanger (1992), Hopenhayn (1992), Hopenhayn and Rogerson (1993).

- The basic goal is to create a model that yields reallocation of labor across establishments (firms) over time
- Households: Representative household with preferences $\sum_{t=0}^{\infty} \beta^t U(c_t, 1 - n_t)$.
- Technology: continuum of establishments (firms) of measure one.
 - Labor is the only input
 - DRS technology: $y_{it} = A_{it}f(n_{it})$, where f is strictly increasing, strictly concave, and A_{it} is stochastic, with transition function Q , the same across establishments but realizations are i.i.d. across establishments
- Aggregate output: $Y_t = \int y_{it} di$, $C_t = Y_t$ (no capital).
- Endowment: each HH has one unit of time in each period, and initial productivities are A_{i0} .
- Markets: in each period, there are markets for output and labor services

State variables

- Individual:
 - HH: none
 - Firm: A
- Aggregate:
 - $\lambda(A)$

(since the only state variable is an exogenous one, no one has a dynamic problem)

RCE definition

A RCE is a list of functions

- $n^c(\lambda)$ • $n^f(A, \lambda)$ • $\pi(\lambda)$
- $c^c(\lambda)$ • $w(\lambda)$ • $H(\lambda)$

such that

(1) (Consumer Maximization)

Taking $w(\lambda), \pi(\lambda)$ as given, for each λ , $c^c(\lambda)$ and $n^c(\lambda)$ solve

$$\begin{aligned} & \max_{c, n} U(c, 1 - n) \\ \text{s.t.} \quad & c = w(\lambda)n + \pi(\lambda) \\ & c \geq 0, \quad 0 \leq n \leq 1 \end{aligned}$$

(2) (Firm Maximization)

Taking $w(\lambda)$ as given, for each A, λ , $n^f(A, \lambda)$ solves

$$\begin{aligned} & \max_n Af(n) - w(\lambda)n \\ \text{s.t.} \quad & n \geq 0 \end{aligned}$$

(3) (Consistency)

- $\pi(\lambda) = \int [Af(n^f(A, \lambda)) - w(\lambda)n^f(A, \lambda)]d\lambda$
- Given λ and Q , the next period's λ is given by $H(\lambda)$

$$\lambda'(\mathbf{A}') = \int_A Q(A, \mathbf{A}')d\lambda,$$

for any Borel set $\mathbf{A}'$ of the space for the state variable A .

(4) (Market Clearing)

$$\begin{aligned} n^c(\lambda) &= \int n^f(A, \lambda)d\lambda \\ c^c &= \int Af(n^f(A, \lambda))d\lambda \end{aligned}$$

for all feasible λ .

A SRCE is a RCE and $\lambda^* : \lambda^* = H(\lambda^*)$.

SRCE algorithm

The algorithm now is easier because we only have an exogenous aggregate state variable. Before we needed the decision rule along with the transition function to find λ^* . Now, λ^* does not depend on the decision rule, so we can solve for the invariant distribution λ^* first.

Given this, we can guess w , solve for the firm's decision rules. Given n^f , find the firm's profits. Aggregate over λ to find the aggregate dividend. Using that, solve the consumer's problem to find n^c . Check market clearing, adjust w accordingly. Notice that we do not need to solve for a new λ^* after each guess of w .

Extension: Model with Capital

- Households: are identical, accumulate capital according to $k' = k(1 - \delta) + i$
- Technology:
 - $y_{it} = A_{it}f(k_{it}, n_{it})$
 - $Y_t = \int y_{it}di, K_t = \int k_{it}di$
- Endowments: 1 unit of time, K_0 units of capital, A_{i0}
- Markets: at each date, there are markets for output, labor and capital

State variables

- Individual:
 - HH: $k \Rightarrow$ dynamic problem because of endogenous state variable
 - Firm: $A \Rightarrow$ static problem, exogenous state variable
- Aggregate:
 - $K, \lambda(A)$

RCE definition

A RCE is a list of functions

- $v(k, K, \lambda)$
- $k^f(A, K, \lambda)$
- $K'(K, \lambda)$
- $k'(k, K, \lambda)$
- $w(K, \lambda)$
- $H(\lambda)$
- $n^c(k, K, \lambda)$
- $r(K, \lambda)$
- $n^f(A, K, \lambda)$
- $\pi(K, \lambda)$

such that

(1) (Consumer Maximization)

Taking $w(K, \lambda)$, $r(K, \lambda)$, $\pi(K, \lambda)$, $K'(K, \lambda)$, $H(\lambda)$ as given, $v(k, K, \lambda)$ solves

$$\begin{aligned}
 v(k, K, \lambda) = & \max_{k', n} \{U(w(K, \lambda)n + r(K, \lambda)k + \pi(K, \lambda) + (1 - \delta)k - k', 1 - n) \\
 & + \beta v(k', K'(K, \lambda), H(\lambda))\} \\
 \text{s.t.} \quad & w(K, \lambda)n + r(K, \lambda)k + \pi(K, \lambda) + (1 - \delta)k - k' \geq 0 \\
 & k' \geq 0, 0 \leq n \leq 1
 \end{aligned}$$

and $n^c(k, K, \lambda)$, $k'(k, K, \lambda)$ are the optimal decision rules

(2) (Firm Maximization)

Taking $w(K, \lambda)$, $r(K, \lambda)$ as given, for each (A, K, λ) , $n^f(A, K, \lambda)$ and $k^f(A, K, \lambda)$ are solutions to

$$\begin{aligned}
 \max_{n, k} \quad & Af(k, n) - w(K, \lambda)n - r(K, \lambda)k \\
 \text{s.t.} \quad & n \geq 0, k \geq 0
 \end{aligned}$$

(3) (Consistency)

- (i) $K = k$
- (ii) $K'(K, \lambda) = k'(K, K, \lambda)$
- (iii) $\pi(K, \lambda) = \int [Af(k^f(A, K, \lambda), n^f(A, K, \lambda)) - r(K, \lambda)k^f(A, K, \lambda) - w(K, \lambda)n^f(A, K, \lambda)] d\lambda$
- (iv) Given λ and Q , the next period's λ is given by $H(\lambda)$

$$\lambda'(\mathbf{A}') = \int_A Q(A, \mathbf{A}') d\lambda,$$

for any Borel set $\mathbf{A}'$ of the set of the state variable A .

(4) (Market Clearing)

For all feasible λ ,

$$K = \int k^f(A, K, \lambda) d\lambda$$

$$n^c(K, K, \lambda) = \int n^f(A, K, \lambda) d\lambda$$

$$w(K, \lambda)n^c(K, K, \lambda) + r(K, \lambda)K + \pi(K, \lambda) + K'(K, \lambda) - K(1 - \delta)$$

$$= \int Af(k^f(A, K, \lambda), n^f(A, K, \lambda)) d\lambda$$

(clears by Walras' Law)

SRCE

A SRCE is a RCE and values K^*, λ^* : $K^* = K'(K^*, \lambda^*)$ and $\lambda^* = H(\lambda^*)$.

As in a standard growth model, in SS we have $\frac{1}{\beta} = 1 + r - \delta$. So given β and δ , we know r^* , and all we need to find is w^* . First find λ^* , then guess w^* , solve for decision rules, and determine aggregates. Then solve consumer's problem and check if the labor market clears. Adjust w^* and iterate.

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